ELEMENTARY AND STRICTLY STANDARD ELEMENTS ARE PRESERVED UNDER ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Note-card: the proof is a definition unfolding through an explicit base-changed presentation; the substitution is written out and is the card's obligation.

Lemma 0.1. *Let $R \to A$ be a ring map of finite presentation. Let $R \to R'$ be a ring map. If $a \in A$ is elementary, resp. strictly standard in A over R, then $a \otimes 1$ is elementary, resp. strictly standard in $A \otimes_R R'$ over R'.*

Proof. If $A = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ is a presentation of A over R, then $A \otimes_R R' = R'[x_1, \ldots, x_n]/(f'_1, \ldots, f'_m)$ is a presentation of $A \otimes_R R'$ over R'. Here f'_j is the image of f_j in $R'[x_1, \ldots, x_n]$. Hence the result follows from the definitions. □

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